Chapter 6

Isomorphic Groups

There is a very natural and important equivalence relation on the class of all groups, that is, a natural sense in which we can view two groups as being "equivalent". Recall that we proved in Chapter 4 that every 7–element group **G** is cyclic, and interpreted this to say that **G** "looks exactly like $\mathbf{Z}_7$." Thus any two 7–element groups look exactly like each other and, in this sense, can be thought of as equivalent. We would like to give a precise meaning to the assertion that two groups "look exactly alike".

As another example, consider the group $\mathbf{K}_4$ and the 4–element subgroup $\mathbf{H} := \{\varnothing, a, c, ac\}$ of the group $\mathbf{P}_{abc}$.

$\mathbf{K}_4$

$*$	E	H	V	D
E	E	H	V	D
H	H	E	D	V
V	V	D	E	H
D	D	V	H	E

$\mathbf{H}$

$\oplus$	$\varnothing$	a	c	ac
$\varnothing$	$\varnothing$	a	c	ac
a	a	$\varnothing$	ac	c
c	c	ac	$\varnothing$	a
ac	ac	c	a	$\varnothing$

We see here is that, although the elements of the two groups are totally different kinds of objects (chess board moves and sets of letters), the tables of the groups look alike. Notice that, if we were to change the order the the elements are listed in the tables, they would no longer look alike. So the equivalence in question has to do with a particular way of pairing the elements:

$$\mathrm{E} \mapsto \varnothing,\ \mathrm{H} \mapsto a,\ \mathrm{V} \mapsto c,\ \mathrm{D} \mapsto ac.$$

Let us call this function h. Then $h : \mathbf{K}_4 \to \mathbf{H}$ is a bijection, that is, a one-to-one and onto function. Moreover, under h the two tables match in the following sense. If X,Y,Z $\in K_4$ and X$*$Y = Z, then h(X)$\bigoplus$ h(Y) = h(Z).

These ideas can be applied to any groups. Let $\mathbf{G}$ be a group with operation $*$ and let $\mathbf{G}'$ be a group with operation $\circ$. A **homomorphism** from $\mathbf{G}$ to $\mathbf{G}'$ is a function $h : \mathbf{G} \to \mathbf{G}'$ with the property that, for all $x, y, z \in \mathbf{G}$,

$$x * y = z \text{ implies } h(x) \circ h(y) = h(z).$$

A function $h : \mathbf{G} \to \mathbf{G}'$ is called an **isomorphism** if it is a homomorphism that is a bijection. We say that groups $\mathbf{G}$ and $\mathbf{G}'$ are **isomorphic** if there is an isomorphism from $\mathbf{G}$ onto $\mathbf{G}'$, and we express this in symbols as $\mathbf{G} \cong \mathbf{G}'$.

Lemma 60. *A function $h : \mathbf{G} \to \mathbf{G}'$ is a homomorphism if and only if, for all $x, y \in \mathbf{G}$, we have*

$$h(x * y) = h(x) \circ h(y).$$

Lemma 30 will be useful for proving the following fact.

Lemma 61. *Two finite cyclic groups of the same order are isomorphic.*

Logarithms were invented by John Napier in order to exploit a special isomorphism between two familiar and important groups:

Lemma 62. *Let $\mathbf{R}^+$ denote the group of all real numbers under addition, and let $\mathbf{R}^\times$ denote the group of all positive real numbers under multiplication. Then $\mathbf{R}^\times \cong \mathbf{R}^+$.*

Homomorphisms, and therefore also isomorphisms, preserve more than just the binary operation.

Theorem 63. *Let $\mathbf{G}$ and $\mathbf{G}'$ be groups with identities $e \in \mathbf{G}$ and $e' \in \mathbf{G}'$, and let $h : \mathbf{G} \to \mathbf{G}'$ be a homomorphism. Then*

(i) $h(e) = e'$,

(ii) $h(x^{-1}) = h(x)^{-1}$ *for each* $x \in \mathbf{G}$,

(iii) *if* $\mathbf{H}$ *is a subgroup of* $\mathbf{G}$, *then* $h(\mathbf{H})$ *is a subgroup of* $\mathbf{G}'$.

(iv) *if* $\mathbf{K}$ *is a subgroup of* $\mathbf{G}'$, *then* $h^{-1}(\mathbf{K}) := \{a \in G \mid h(a) \in \mathbf{K}\}$ *is a subgroup of* $\mathbf{G}$.

As a special case of part (iv), the subgroup $h^{-1}(\{e'\})$ of $\mathbf{G}$ is called the **kernel** of h and is denoted by $\mathbf{K}_h$. This subgroup is useful for recognizing when a homomorphism is an isomorphism. Applying part (i) we see that, if h happens to be one-to-one, then $\mathbf{K}_h = \{e\}$. The converse is also true.

Lemma 64. *Let $h : \mathbf{G} \to \mathbf{G}'$ be a homomorphism. Then h is one-to-one if and only if $\mathbf{K}_h = \{e\}$.*

Theorem 65. *Isomorphism is an equivalence relation on the class of all groups, that is, for all groups $\mathbf{G}$, $\mathbf{G}'$ and $\mathbf{G}''$,*

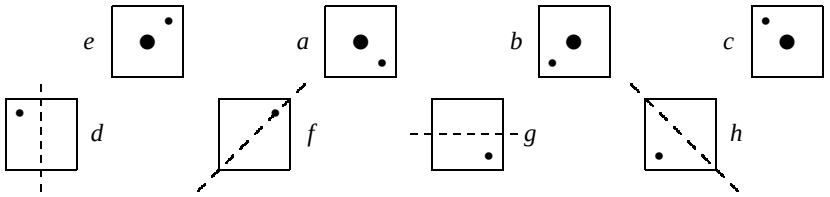
(i) $\mathbf{G} \cong \mathbf{G}$,

(ii) $\mathbf{G} \cong \mathbf{G}'$ *implies* $\mathbf{G}' \cong \mathbf{G}$, *and*

(iii) $\mathbf{G} \cong \mathbf{G}'$ *and* $\mathbf{G}' \cong \mathbf{G}''$ *implies* $\mathbf{G} \cong \mathbf{G}''$.

...

We will now digress briefly to present an important class of finite non-commutative groups. For a positive integer $n \geq 3$, the **dihedral n-group $\mathbf{D}_n$** consists of the rigid motions of a regular n-sided polygon. Since the regular n-gon has n top side orientations and n bottom side orientations, the order of $\mathbf{D}_n$ is $2n$. In case $n = 3$, the group $\mathbf{D}_3$ is the familiar 6-element group $\mathbf{S}_3$. The 8-element dihedral 4-group, $\mathbf{D}_4$, is illustrated below:



Notice that the 4 top side orientations are obtained by successive rotations of the square, while the 4 bottom side orientations can be realized by flip d followed by successive clockwise rotations. In algebraic language, this says that the set $\{a,d\}$ **generates $\mathbf{D}_4$**, as we have

$$b = a^2,\ c = a^3,\ e = a^4;\ f = ad,\ g = a^2d,\ h = a^3d.$$

Instead of listing the elements of $\mathbf{D}_4$ as

$$\mathbf{D}_4 = \{e,a,b,c,d,f,g,h\},$$

we find that the listing

$$\mathbf{D}_4 = \{e,a,a^2,a^3,d,ad,a^2d,a^3d\}$$

provides much more informative names for the elements.

In fact, these names allow us to give a complete description of the group if we remember a small bit of additional information. Clearly, a has order 4 and d has order 2. Going back to the square itself, we can compute the product $da = h = a^3d$. Thus we can describe $\mathbf{D}_4$ by giving **generators** and **defining relations** as

$$\mathbf{D}_4 = \mathbf{sg}\{a,d \mid a^4 = e = d^2,\ da = a^3d\}. \qquad (\star)$$

This means that $\mathbf{D}_4$ consists of all products of a's and d's subject only to the constraints implied by the defining relations. From this information we can deduce that $\mathbf{D}_4$ has the 8 elements listed, and we can fill in the complete table for $\mathbf{D}_4$. For example, to compute the product of a^3d and a^2d, we have

$$(a^3d)(a^2d) = a^3(da)ad = a^3(a^3d)ad = a^6(da)d$$
$$= a^6(a^3d)d = a^9d^2 = a(a^4)(a^4)(d^2) = aeee = a$$

The strategy here is to use $da = a^3d$ to express the product as a power of a times a power of d, and then to use $a^4 = e = d^2$ to reduce the exponents. Similarly, for any positive integer n, we can describe $\mathbf{D}_n$ as

$$D_n = \mathbf{sg}\{a,d \mid a^n = e = d^2,\ da = a^{n-1}d\}.$$

where a has order n and b as order 2, and $ba = a^{n-1}b$.

Problem 66. *Using $(\star)$, fill out a table for $\mathbf{D}_4$ without referring to the square itself.*

Problem 67. *Using the description*

$$\mathbf{S}_3 = \mathbf{sg}\{a, c \mid a^3 = e = c^2,\ ca = a^2c\},$$

list the distinct elements of $\mathbf{S}_3$ and fill out its table without referring to the triangle.

...

The notion of isomorphism allows us to articulate a fundamental goal of group theory. Given a property P that a group may or may not have, we seek a **representation theorem** for this property. What this means is that we would like to find a set of specific well understood groups that have property P, and then prove that every group with property P is isomorphic to a group in our set. For example, let P be the property of having prime order p. A well understood group of order p is the group $\mathbf{Z}_p$. If $\mathbf{G}$ has property P, then $\mathbf{G}$ is cyclic by Theorem 47 and consequently $\mathbf{G} \cong \mathbf{Z}_p$ by Lemma 61. This gives us a nice representation theorem.

Theorem *Every group of prime order p is isomorphic to $\mathbf{Z}_p$.*

In order to prove the following representation theorems, you will need to spend some time examining elements of the group, considering their possible orders, and looking at the subgroups they generate. Recall that, if p and q are both prime, then the only divisors of pq are $1, p, q$ and pq.

Theorem 68. *Let p and q be distinct primes. Then every commutative group of order pq is cyclic and is therefore isomorphic to $\mathbf{Z}_{pq}$.*

Lemma 69. *Let p be a prime. Then every commutative group of order p^2 is isomorphic to either the cyclic group $\mathbf{Z}_{p^2}$ or to the direct product $\mathbf{Z}_p \times \mathbf{Z}_p$.*

It turns out that there is a beautiful representation theory for all finite *commutative* groups which extends the three previous results. The final theorem, which we won't prove here, states that *every finite commutative group is isomorphic to a direct product of cyclic groups*.

Representation of non-commutative groups is a fascinating subject that proves to be much more difficult. In fact, a full description of all finite groups is not yet known. A well known representation theorem for all groups was first discovered by Sir Arthur Cayley (1821-1895). In Theorem 15 we saw that the left multiplication function ℓ associates each element a of a group $\mathbf{G}$ with a permutation ℓ_a of $\mathbf{G}$. Thus $\ell : \mathbf{G} \to \mathbf{S}_\mathbf{G}$. In fact, ℓ is a one-to-one homomorphism:

Cayley Representation Theorem 70. *Every group $\mathbf{G}$ is isomorphic to a subgroup of the permutation group $\mathbf{S}_X$ for some set X, namely, for $X = \mathbf{G}$.*

In the remainder of this chapter we will prove two important theorems which, together with what we have, will provide a full description of finite groups of many small orders. Our next theorem extends Theorem 68 to all groups in the case that $p = 2$.

Theorem 71. *Let q be a prime. Then every group of order $2q$ is isomorphic to either the cyclic group $\mathbf{Z}_{2q}$ or to the dihedral group $\mathbf{D}_q$.*

The Lemma 69 is called a "lemma" because, as we will now see, it can be extended to all groups by utilizing some new techniques. Elements a and b of a group $\mathbf{G}$ are said to be **conjugate** if there is an element $x \in \mathbf{G}$ such that $b = x^{-1}ax$. In this case we write $a \equiv_c b$ and say that a **is conjugate to** b.

Theorem 72. *For a group* **G**, *the relation* $\equiv_c$ *is an equivalence relation.*

Theorem 73. *For a group* **G**,

(i) *Conjugacy is the equality relation if and only if* **G** *is commutative.*

(ii) *Conjugate elements of* **G** *have the same order.*

Problem 74. *Find all of the conjugate classes of* $\mathbf{S}_3$.

The $\equiv_c$–classes of **G** are called the **conjugate classes** of **G**.

If a is an element of the group **G**, the **normalizer** of a is defined to be the set

$$\mathbf{N}_a := \{x \in \mathbf{G} \mid xa = ax\}$$

of elements that commute with a.

Lemma 75. *The normalizer of an element of a group is a subgroup.*

Lemma 76. *Let* **G** *be a group with* $a,x,y \in \mathbf{G}$. *Then* $\mathbf{N}_a x = \mathbf{N}_a y$ *if and only if* $x^{-1}ax = y^{-1}ay$. [*In words, two elements are in the same right coset of the normalizer of a if and only if they produce the same conjugate of a*.]

Recall that if **H** is a subgroup of a finite group **G**, then the **index** of **H** in **G** is the number $[\mathbf{G} : \mathbf{H}]$ of right (or left) cosets of **H** in **G**. LaGrange's Theorem now tells us something important about conjugate classes.

Lemma 77. *Let* **G** *be a finite group with* $a \in \mathbf{G}$. *Then the order of the conjugate class of a is equal to the index of the normalizer of a, that is,* $\circ([a]_{\equiv_c}) = [\mathbf{G} : \mathbf{N}_a]$. *In particular,* $\circ([a]_{\equiv_c}) | \circ(\mathbf{G})$.

(Show that the set of conjugates of a and the set of right cosets of $\mathbf{N}_a$ are the *same size*.)

A subgroup always contains at least one element, the identity, and the conjugate class of a always contains at least the one element a. If one of these sets contains only this one element it is said to be **trivial**; otherwise it is **non-trivial**. The **center** of a group **G** is defined as

$$Z(\mathbf{G}) := \{b \in \mathbf{G} \mid ba = ab \text{ for all } a \in \mathbf{G}\},$$

the elements b of **G** that commute with every element of **G**. In other words, $Z(\mathbf{G})$ in the intersection of all of the normalizers of elements of **G**.

Lemma 78. *The center* $Z(\mathbf{G})$ *is a subgroup of* **G** *which is exactly the union of the trivial conjugate classes of G.*

Now let **G** be a finite group and let $a_1, a_2, \ldots, a_m \in \mathbf{G}$ be a list of representatives of the distinct non-trivial conjugate classes of **G**. Then we can partition **G** into its trivial and non-trivial conjugate classes as

$$\mathbf{G} = Z(\mathbf{G}) \cup [a_1]_{\equiv_c} \cup [a_2]_{\equiv_c} \cup \cdots \cup [a_m]_{\equiv_c}.$$

It follows that the order of **G** is the sum of the number of elements in each piece of this partition. This assertion yields an important numerical property of a group that is called the **Class Equation** of **G**:

$$\circ(\mathbf{G}) = \circ(Z(\mathbf{G})) + \circ([a_1]_{\equiv_c}) + \circ([a_2]_{\equiv_c}) + \cdots + \circ([a_m]_{\equiv_c}).$$

Lemma 79. *If the order of a group is a power of a prime, then the group has a non-trivial center.*

Theorem 80. *Let p be a prime. Then every group of order p^2 is commutative, and is therefore isomorphic to either the cyclic group $\mathbf{Z}_{p^2}$ or to the direct product $\mathbf{Z}_p \times \mathbf{Z}_p$.*

Problem 81. *For which of the numbers n from* 1 *to* 25 *do you now know, up to isomorphism, all of the groups of order n? Make a list which gives, for each such number n, all of the non-isomorphic n–element groups.*